Mock Exam – Digital Health & AI in Healthcare (WS 2025/2026)

Total: 52 points
Time: 60 minutes

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1. AI Methods & Clinical Decision Support (14 points)

1.1 Matching Algorithms to Data Type and Reason (4 points)

Match each AI algorithm to the most appropriate:

- **Data type (List A)**
- **Reason (List B)**

List A – Data Types

A1. Discrete or categorical clinical outcomes
A2. High-dimensional imaging or omics data
A3. Small sample size datasets
A4. Longitudinal or time-series patient data
A5. Probabilistic relationships among variables
A6. Structured tabular EHR data

List B – Reasons

B1. Captures long-term dependencies with memory cells
B2. Encodes uncertainty via conditional probability distributions
B3. Learns compressed latent representations
B4. Performs well using prior assumptions to improve sample efficiency
B5. Models categorical likelihoods for classification
B6. Handles mixed numerical and categorical predictors efficiently

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Algorithms

1. LSTM
2. Bayesian Network
3. Variational Autoencoder
4. Naïve Bayes

Answer:

1.2 LSTM Data Table Completion (4 points)

You are given hourly glucose measurements processed for an LSTM model.

Rule:
Hyperglycemia = 1 if (Glucose ≥ 180 AND Mask = 1), else 0

Δt = time since last valid measurement (Mask = 1).

Time	Glucose	Mask	Δt	Hyperglycemia
08:00	165	1	0	[A]
09:00	NaN	0	[B]	0
10:00	190	1	0	[C]
11:00	NaN	0	[D]	0
12:00	NaN	0	[E]	0
13:00	210	1	0	[F]

Fill in A–F.

1.3 Evaluation Metric Selection (3 points)

For each scenario, select the **most appropriate metric** and briefly justify (1 sentence each).

Metrics to choose from:

- Accuracy
- Recall (Sensitivity)
- Precision (PPV)
- F1 Score
- AUC-ROC
- AUPRC

a) Screening tool for a rare but fatal disease

b) Confirmatory test before prescribing a toxic treatment

c) Comparing two ML models across all classification thresholds

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1.4 Short Answer (3 points)

a) Name two challenges of AI in clinical decision support systems.

b) What is the difference between overfitting and underfitting?

c) Why are external validation cohorts important in medical AI?

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2. Digital Biomarkers (8 points)

2.1 Time-Series in Healthcare (2 points)

List four examples of time-series data in healthcare.

2.2 Traditional vs Digital Endpoints (3 points)

Complete the table by assigning each item to:

- Endpoint
- Clinical Outcome
- Biomarker

Item	Category
Wearable-measured step count variability	
HbA1c blood test	
Patient-reported depression questionnaire score	
Continuous heart rate variability from smartwatch	
MRI-based tumor volume	

2.3 Mental Health Digital Biomarkers (3 points)

Name three potential digital biomarkers for depression or anxiety derived from smartphone or wearable data and briefly state what behavioral signal they reflect.

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3. Digital Twin (14 points)

3.1 Conceptual Questions (4 points)

a) Define a Digital Twin in healthcare.

b) What is the difference between a mechanistic model and a data-driven model?

c) Why is population scaling important in Digital Twin models?

d) Name one limitation of Digital Twins in clinical practice.

3.2 Agent-Based Model Components (4 points)

Match each category to one definition (List A) and one example (List B).
Do not reuse items.

List A – Definitions

A1. Autonomous entities with behavioral rules
A2. Spatial or network structure where interactions occur
A3. System-wide parameters shared by all entities
A4. Emergent population-level outcome

List B – Examples

B1. Individual vaccination status
B2. Transmission rate parameter
B3. Social contact network graph
B4. Epidemic curve over time

Category	Definition	Example
Agent		
Environment		
Global Variable		
Emergent Output		

3.3 DE vs ABM Model Choice (3 points)

For each scenario, choose:

- Differential Equation (DE) Model
- Agent-Based Model (ABM)

Provide one concise reason.

a) Modeling national-level influenza prevalence trends.

b) Studying infection spread in a small school with heterogeneous contact patterns.

3.4 SEIR Model Matching (3 points)

Match each compartment (S, E, I, R) to:

- A description (List A)
- A governing parameter (List B)

Do not reuse descriptions.

List A – Descriptions

A1. Infected and infectious individuals
A2. No prior exposure, susceptible to infection
A3. Infected but not yet infectious (latent)
A4. Recovered with immunity

List B – Parameters

B1. Transmission rate (β)
B2. Incubation rate (σ)
B3. Recovery rate (γ)

Answer:

4. Model-Guided Medicine (8 points)

4.1 Bayesian Networks (4 points)

a) What is a Bayesian Network?

b) What does an edge represent?

c) Why are Bayesian Networks useful in clinical decision-making?

d) What is meant by inference under uncertainty?

4.2 Hierarchical Models (2 points)

A statistical model that organizes parameters at multiple levels (e.g., patient, hospital, region) is called a _____ model.

Explain in one sentence why such models are useful in healthcare data.

4.3 Wisdom-Oriented Healthcare (2 points)

What distinguishes wisdom-oriented healthcare from purely data-driven prediction systems?

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5. Infrastructure, Standards & Regulations (8 points)

5.1 IT Infrastructure (3 points)

Name three components required for implementing an AI-based decision support system in a hospital IT environment.

5.2 DiGA and Regulation (3 points)

a) What is a DiGA?

b) Name two requirements for regulatory approval of digital health applications in Europe.

c) Why is evidence generation important for reimbursement?

5.3 Data Governance (2 points)

a) What is the purpose of anonymization or pseudonymization in medical datasets?

b) Why is interoperability (e.g., HL7 FHIR) important for digital health systems?